7.2 二重积分的计算下

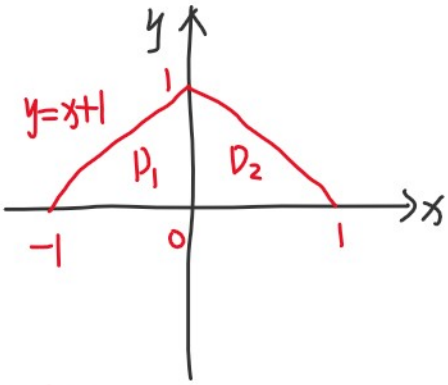
2022年3月31日 16:42

$D_2: xy \geq 0$

$D_1: xy \leq 0$

= 找 | | = 0的线分开.

21. 计算$\iint\limits_D |xy| \, d\sigma$，其中 D 是由 x 轴，$y+x=1$ 和 $y-x=1$ 围成的闭区域.



解 如图. $\iint\limits_D |xy| \, d\sigma$

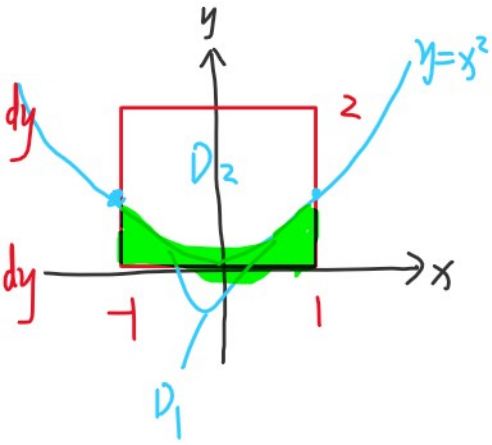
$$= \iint\limits_{D_1} (-xy) \, dx \, dy + \iint\limits_{D_2} xy \, dx \, dy$$

$$= \int_{-1}^{0} dx \int_{0}^{x+1} (-xy) \, dy + \int_{0}^{1} dx \int_{0}^{1-x} xy \, dy$$

$$= 1/12.$$

22. 设 $D = \{(x,y) \mid -1 \leqslant x \leqslant 1, 0 \leqslant y \leqslant 2\}$，求$\iint\limits_D \sqrt{|y-x^2|} \, dx dy$.

$y \geq x^2$



$y \leq x^2$

如图，原积分 $= \iint\limits_{D_1} \sqrt{|y-x^2|} \, dx dy + \iint\limits_{D_2} \sqrt{|y-x^2|} \, dx dy$

$$= \int_{-1}^{1} dx \int_{0}^{x^2} \sqrt{x^2-y} \, dy + \int_{-1}^{1} dx \int_{x^2}^{2} \sqrt{y-x^2} \, dy$$

$$= \int_{-1}^{1} \frac{2}{3} x^3 \, dx + \frac{2}{3} \int_{-1}^{1} (2-x^2)^{\frac{3}{2}} \, dx$$

$$= \frac{2}{3}\cdot 2\int_0^1 \left(\sqrt{2-x^2}\right)^3 dx$$

令 $x=\sqrt{2}\sin t$, $\quad x=0,\ t=0$

$\qquad\qquad\qquad\qquad x=1,\ t=\frac{\pi}{4}$

$dx = d\sqrt{2}\sin t = \sqrt{2}\cos t\, dt$

原积分 $= \frac{4}{3}\int_0^{\frac{\pi}{4}} 2\sqrt{2}\cdot\cos^3 t\cdot\sqrt{2}\cos t\, dt$

$$= \frac{16}{3}\int_0^{\frac{\pi}{4}} \cos^4 t\, dt \qquad\qquad \int \cos^2 t$$

$$= \frac{16}{3}\int_0^{\frac{\pi}{4}} \left(\frac{1+\cos 2t}{2}\right)^2 dt \qquad\qquad \cos^2 t = \frac{1+\cos 2t}{2}$$

$$= \frac{16}{3}\cdot\frac{1}{4}\int_0^{\frac{\pi}{4}} (1+\cos 2t)^2 dt$$

$$= \frac{4}{3}\int_0^{\frac{\pi}{4}} \left(1+\underline{\cos^2 2t}+2\cos 2t\right) dt.$$

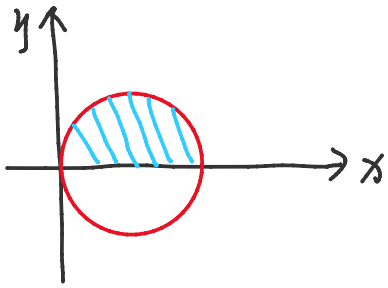
$$= \frac{\pi}{2}+\frac{4}{3}$$

$$\int_0^{\frac{\pi}{4}} \cos^2 2t\, dt \xlongequal{2t=u} \frac{1}{2}\int_0^{\frac{\pi}{2}} \cos^2 u\, du$$

$$= \frac{1}{2}\cdot\frac{1}{2}\cdot\frac{\pi}{2} = \frac{\pi}{8}$$

27. 计算$\iint\limits_D \sqrt{4-x^2-y^2}\,d\sigma$，其中 D 是由 Ox 轴及曲线 $y=\sqrt{2x-x^2}$ 所围成的闭区域.

原积分 $= \int_0^{\frac{\pi}{2}} d\theta \int_0^{2\cos\theta} \sqrt{4-r^2}\cdot r\,dr$



$(1,0)$ $r=1$

$= \int_0^{\frac{\pi}{2}} -\frac{1}{3}(8\sin^3\theta-8)\,d\theta$

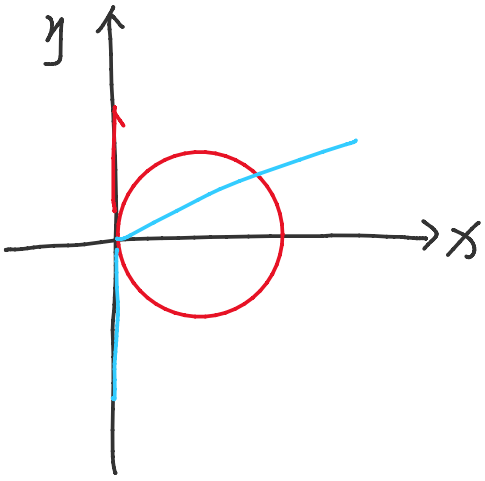
$= -\frac{8}{3}\int_0^{\frac{\pi}{2}}(\sin^3\theta-1)\,d\theta.$

$= -\frac{8}{3}\left(\frac{2}{3}-\frac{\pi}{2}\right)$

$= -\frac{16}{9}+\frac{4}{3}\pi$

28. 计算$\iint\limits_D \sqrt{x}\,dx\,dy$，其中 $D: x^2+y^2 \leqslant x$.

$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{\cos\theta} \sqrt{r\cos\theta}\; r\,dr$

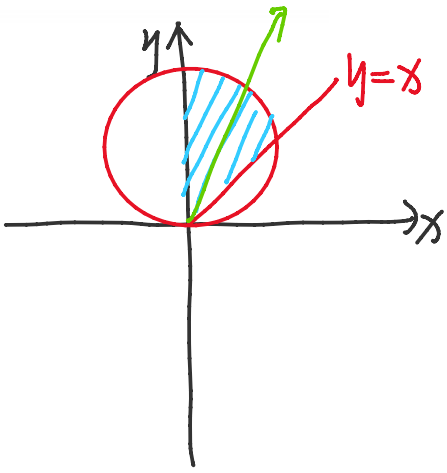


$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{2}{5}\cos^3\theta\,d\theta$

$$= 2\cdot\frac{2}{5}\int_0^{\frac{\pi}{2}}\cos^3\theta\,d\theta$$

$$= \frac{4}{5}\cdot\frac{2}{3} = \frac{8}{15}$$

30. 设区域 D 是由不等式组 $y \geqslant x, x^2+y^2 \leqslant 2y, x \geqslant 0$ 所确定的闭区域，求二重积分 $\iint\limits_D xy\,d\sigma$.



$$\text{原积分} = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \int_0^{2\sin\theta} r^2\cos\theta\sin\theta\, r\,dr$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} 4\sin^5\theta\cdot\cos\theta\,d\theta$$

$$= 4\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \sin^5\theta\, d\sin\theta$$

$$= 4\cdot\frac{1}{6}\sin^6\theta\Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= \frac{2}{3}\left(1-\frac{1}{8}\right)$$

$$= \frac{2}{3}\cdot\frac{7}{8} = \frac{7}{12}$$

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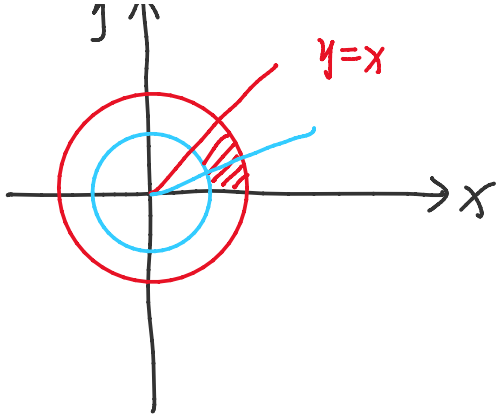
32. 计算 $\iint\limits_D \frac{y}{x}dxdy$，其中 D 是由圆周 $x^2+y^2=4$，$x^2+y^2=1$，直线 $y=0$ 及 $y=x$ 所围成的在第一象限内的闭区域.

原积分 $= \int_0^{\frac{\pi}{4}} d\theta \int_1^2 \tan\theta\, r\,dr$

$$原积分 = \int_0^{\frac{\pi}{4}} d\theta \int_1^2 \tan\theta \; r\,dr$$



$$= \frac{3}{2}\int_0^{\frac{\pi}{4}} \tan\theta \, d\theta$$

$$= -\frac{3}{2}\ln|\cos\theta|\Big|_0^{\frac{\pi}{4}}$$

$$= -\frac{3}{2}\left(\ln\frac{1}{\sqrt{2}} - 0\right)$$

$$= \frac{3}{2}\ln\sqrt{2} = \frac{3}{2}\cdot\frac{1}{2}\ln 2$$

$$= \frac{3}{4}\ln 2$$

19. 若 $\lim\limits_{x\to1} f(x)$ 存在，且 $f(x) = x^3 + \frac{2x^2+1}{x+1} + 2\lim\limits_{x\to1} f(x)$，则 $\lim\limits_{x\to1} f(x) =$ _____.

a

$a = \lim\limits_{x\to1} f(x) = x^3 + \boxed{} + 2a$ 8页

14. 设函数 $f(x)$ 满足 $f(x) = 4x^3 + 2x + 3\lim\limits_{x\to1} f(x)$，求 $\int f(x)\,dx$. —44—14

$f(x) = 4x^3 + 2x + 3a$ a

19. 若 $f(x) = \frac{1}{1+x^2} + \sqrt{1-x^2}\int_0^1 f(t)\,dt$，求 $f(x)$.

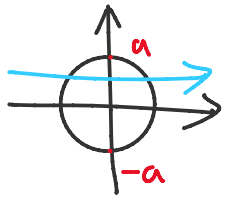
a

$a = \int_0^1 f(x)dx = \int_0^1\left(\frac{1}{1+x^2} + a\sqrt{1-x^2}\right)dx$

34. 设 $f(x,y)$ 为连续函数，且 $f(x,y) = x\iint\limits_{x^2+y^2\leqslant a^2} f(x,y)\,dx\,dy + y^2$，其中 $a>0$，求 $f(x,y)$.

a

解. 设 $\iint\limits_{x^2+y^2\le a^2} f(x,y)\,dx\,dy = a.$

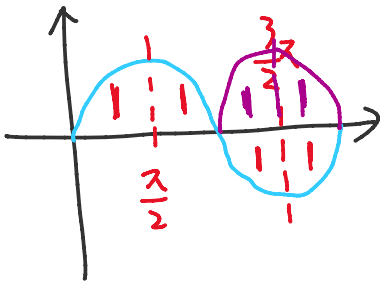


则 $f(x,y) = ax + y^2$

则 $\iint\limits_{x^2+y^2\le a^2} f(x,y)\,dx\,dy = \iint\limits_{x^2+y^2\le a^2} (ax+y^2)\,dx\,dy$

则 $a = 0 + \iint\limits_{x^2+y^2\le a^2} y^2\,dx\,dy$

$= \int_0^{2\pi} d\theta \int_0^a r^2\cdot\sin^2\theta \; r\,dr$



$= \frac{1}{4}a^4 \int_0^{2\pi} \sin^2\theta\,d\theta$

$= \frac{1}{4}a^4 \cdot 4 \int_0^{\frac{\pi}{2}} \sin^2\theta\,d\theta.$

$= a^4 \cdot \frac{1}{2}\cdot\frac{\pi}{2} = \frac{1}{4}a^4\pi.$

则 $f(x,y) = \frac{\pi}{4}a^4 x + y^2.$

35. 设区域 $D: x^2+y^2 \leqslant a^2 (a>0)$. $f(xy, x+y) = \begin{cases} x^2+y^2+xy, & x \geqslant 0, y \geqslant 0, \\ 0, & \text{其他}, \end{cases}$

(1) 求 $f(x,y)$;

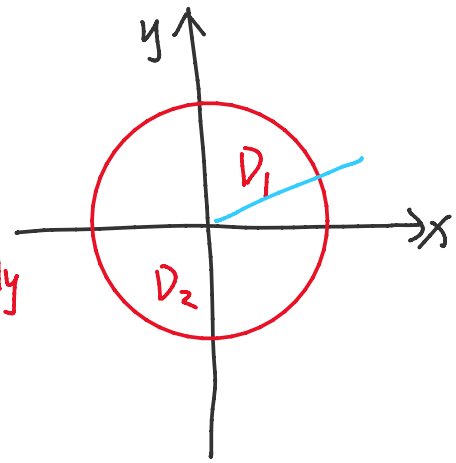
(2) 计算 $\iint\limits_D f(x,y)\mathrm{d}x\mathrm{d}y$.

$x \geqslant 0, y \geqslant 0$ 时, $f(xy, x+y) = x^2+y^2+xy$

$= (x+y)^2 - xy$

则 $x \geqslant 0, y \geqslant 0$ 时, $f(x,y) = y^2 - x$.

则 $f(x,y) = \begin{cases} y^2-x, & x \geqslant 0, y \geqslant 0 \\ 0, & \text{其他}. \end{cases}$



(2) $\iint\limits_D f(x,y)\,dx\,dy = \iint\limits_{D_1} f(x,y)\,dx\,dy + \iint\limits_{D_2} f(x,y)\,dx\,dy$

$= \int_0^{\frac{\pi}{2}} d\theta \int_0^a (r^2\sin^2\theta - r\cos\theta)\, r\,dr$

$= \int_0^{\frac{\pi}{2}} \left(\frac{1}{4}a^4 \sin^2\theta - \frac{1}{3}a^3\cos\theta\right) d\theta$

$= \frac{1}{4}a^4 \cdot \frac{\pi}{4} - \frac{1}{3}a^3 \cdot 1$

$\int_0^{\frac{\pi}{2}} \sin^2\theta\, d\theta$

$= \frac{1}{2} \cdot \frac{\pi}{2}$

$= \frac{1}{16}a^4\pi - \frac{a^3}{3}$